



ROI

Milestone Test-01

Phase-02

DURATION: 180 Minutes

DATE: 08/06/2025

M.MARKS: 300

ANSWER KEY

PHYSICS

1. (1)
2. (3)
3. (2)
4. (3)
5. (2)
6. (1)
7. (1)
8. (1)
9. (1)
10. (2)
11. (2)
12. (2)
13. (2)
14. (4)
15. (4)
16. (1)
17. (4)
18. (2)
19. (2)
20. (4)
21. (1)
22. (4)
23. (300)
24. (32)
25. (2)

CHEMISTRY

26. (3)
27. (2)
28. (3)
29. (2)
30. (2)
31. (1)
32. (1)
33. (1)
34. (3)
35. (1)
36. (2)
37. (1)
38. (3)
39. (3)
40. (4)
41. (1)
42. (3)
43. (2)
44. (4)
45. (1)
46. (2)
47. (4)
48. (200)
49. (75)
50. (1)

MATHEMATICS

51. (1)
52. (2)
53. (1)
54. (1)
55. (3)
56. (2)
57. (3)
58. (3)
59. (2)
60. (2)
61. (1)
62. (1)
63. (2)
64. (4)
65. (2)
66. (2)
67. (4)
68. (3)
69. (4)
70. (1)
71. (1)
72. (0)
73. (0)
74. (1)
75. (1)

SECTION-I (PHYSICS)

1. (1)

Equal charges at all three corners

Symmetry ensures fields cancel at centroid.

2. (3)

Here, $\theta = 30^\circ$, $E = 10^4 \text{ NC}^{-1}$,

$\tau = 8 \text{ N-m}$, $q = ?$, $2a = 4 \text{ cm} = 4 \times 10^{-2} \text{ m}$

From, $\tau = pE \sin \theta = q(2a)E \sin \theta$

$$\begin{aligned} \text{or } q &= \frac{\tau}{2aE \sin \theta} \\ &= \frac{8}{4 \times 10^{-2} \times 10^4 \times \sin 30^\circ} \\ &= \frac{8 \times 2}{4 \times 10^2} \end{aligned}$$

or $q = 4 \times 10^{-2} \text{ C}$

3. (2)

$$F = \frac{kq(6-q)}{d^2}$$

$$\frac{dF}{dq} = 0$$

$$\Rightarrow \frac{k}{d^2} (6q - q^2) = 0$$

$$q = 3 \mu\text{C}$$

4. (3)

Electric field

$$E = \frac{KQx}{(x^2 + R^2)^{\frac{3}{2}}}$$

For maximum $\frac{dE}{dx} = 0$

After solving we get, $\left(x = \pm \frac{R}{\sqrt{2}} \right)$

5. (2)

$$\tau_{\max} = pE \sin 90^\circ$$

$$= q(a)E = 2 \times 10^{-6} \times 4 \times 10^{-2} \times 10^5$$

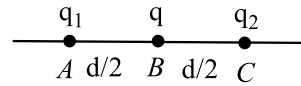
$$= 8 \times 10^{-3} \text{ Nm}$$

6. (1)

Symmetry ensures cancellation of field vectors.

7. (1)

For equilibrium, we have



$$\vec{F}_{AB} + \vec{F}_{AC} = 0$$

$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{q_1 q}{(d/2)^2} + \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{d^2} = 0$$

Given, $q_1 = q_2 = -1 \mu\text{C}$

$$\text{So, } -\frac{q}{(d/2)^2} + \frac{1}{d^2} = 0 \text{ or } q = \frac{1}{4} = 0.25 \text{ C}$$

8. (1)

As the electron is moving in a straight line and starts from rest.

Thus, from

$$s = ut + \frac{1}{2} at^2$$

We get

$$h = (0)t + \frac{1}{2} a_e t^2$$

$$t = \sqrt{\frac{2h}{a_e}}$$

$$= \sqrt{\frac{2hm_e}{eE}} \quad \left(a_e = \frac{eE}{m_e} \right)$$

$$= \sqrt{\frac{2 \times 1.5 \times 10^{-2} \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19} \times 2 \times 10^4}} = 2.92 \times 10^{-9} \text{ s}$$

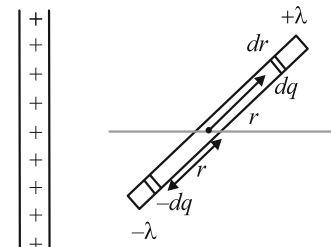
9. (1)

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{d^2}$$

$$\therefore 40 = \frac{(9 \times 10^9) \times q^2}{(0.6)^2}$$

$$\Rightarrow q = 40$$

10. (2)



E.F. is constant

$$E = \frac{\sigma}{2\epsilon_0} \quad \dots (1)$$

Let a small dipole of dq charge at r distance from O, so Torque

$$d\tau = dPE \sin \theta$$

$$d\tau = dq(2r)E \sin \theta, \quad dq = \lambda dr$$

$$\Rightarrow \tau = \int_0^{\ell} \lambda(2r)E \sin \theta dr = 2\lambda E \sin \theta \left[\frac{r^2}{2} \right]_0^{\ell}$$

$$= \lambda E \sin \theta \ell^2 = \frac{\lambda \sigma \ell^2 \sin \theta}{2\epsilon_0} = \frac{\lambda \sigma \ell^2}{4\epsilon_0}$$

11. (2)

Electric field on axis is:

$$E = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + R^2)^{3/2}}$$

12. (2)

In case of an electric dipole, $F \propto \frac{1}{r^3}$

$\therefore$ new force $= F/2^3 = F/8$

13. (2)

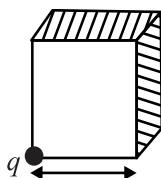
- (a) E_C and E_F are cancelled. E_E and E_D at 60°
 (b) E_B and E_E are cancelled. E_F and E_D at 120° .
 (c) E_B and E_E are cancelled. Similar, E_F and E_C are cancelled.
 (d) E_F and E_D at 120° . So, their resultant is in the direction of E_E . Hence, net is $2E$.

14. (4)

Total flux coming out from unit charge

$$\phi = \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} = \epsilon_0^{-1}$$

15. (4)



Zero flux passes through the 3 surfaces which are connected to the point charge, so by symmetry, equal flux passes through the remaining three surfaces of the cube. Total flux through the cube is $\left(\frac{1}{8}\right)^{\text{th}}$ of the total flux (q/ϵ_0) through the point charge. So, flux through one surface $= \frac{2}{3} \frac{q}{8\epsilon_0} = \frac{q}{12\epsilon_0}$

16. (1)

Dipole moment

$$p = 2 \times 10^{-8} \times 2 \times 10^{-4} = 4 \times 10^{-12} \text{ Cm}$$

$$\text{Maximum torque} = pE = 4 \times 10^{-12} \times 4 \times 10^8 = 16 \times 10^{-4} \text{ Nm}$$

17. (4)

At mid point, $E = 0$

Before mid point, E is positive. This is maximum near the charge and decreases towards mid point

After mid point, E is negative. The curve crosses x -axis at $x = d/2$. From centre to end, E decreases

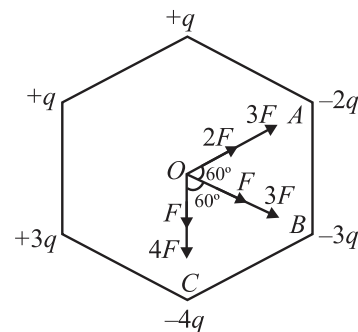
The variation is shown by curve in option (4)

18. (2)

In below figure, we see that force along $OA = 5F$ and force along $OC = 5F$,

Therefore, by symmetry, net force will be along OB , and components perpendicular to OB will be cancelled.

Hence $F_{\text{net}} = 5F \cos 60^\circ + 5F \cos 60^\circ + 4F = 9F$ along OB



19. (2)

E.F. will be parallel to the tangent to the nearby field lines

20. (4)

Force due to adjacent:

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{a^2}$$

Two adjacent are perpendicular $\rightarrow$ resultant $\sqrt{2}F$.

Diagonal charge force:

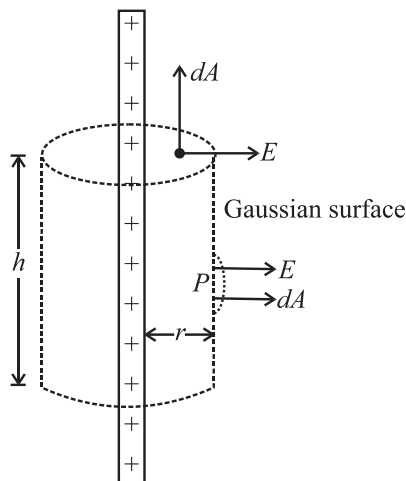
$$F_d = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{2a^2}$$

Adding components,

$$F_{\text{net}} = \frac{Q^2}{4\pi\epsilon_0 a^2} \left(\sqrt{2} + \frac{1}{2} \right)$$

21. (1)

According to Gauss theorem for closed surface



Taking cylindrical Gaussian surface of radius r , height h , curved surface area = $2\pi rh$

Electric flux through it = $E \times 2\pi rh$

Charge enclosed = λh

From Gauss theorem,

Charge = $\epsilon_0 \times$ (Electric flux)

$$\Rightarrow \epsilon_0 E \times 2\pi rh = \lambda h$$

$$\Rightarrow E = \frac{1}{2\pi\epsilon_0} \cdot \frac{\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r}$$

22. (4)

For equilibrium net force on q must be zero

$$i.e. |F_1| = |F_2|$$

$$\Rightarrow x = 4\text{cm}$$

23. (300)

Angle between two forces due to individual charges is equal to 60°

$$\therefore R = \sqrt{F^2 + F^2 + 2F \cdot F \cos 60^\circ} = F\sqrt{3}$$

$$= 10\sqrt{3}$$

$$= \sqrt{300}N$$

24. (32)

Force on proton at point A,

$$F_A = qE_A = -1.6 \times 10^{-19} \times 20$$

$$= -32 \times 10^{-19}$$

25 (2)

Ratio of charges = 1:2

$$\therefore q_1 = \frac{1}{3} \times 1\mu\text{C} \quad \text{and} \quad q_2 = \frac{2}{3} \times 1\mu\text{C}$$

Electrostatic force between the two charges

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$= \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 1 \times 10^{-6}}{3 \times 3 \times (1)^2}$$

$$= 2 \times 10^{-3}$$

SECTION-II (CHEMISTRY)

26. (3)

Both the molecules are polar and possess dipole.

27. (2)

Normality of the mixed solution

$$= \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2}$$

$$= \frac{0.6 \times \frac{100}{1000} + 0.3 \times \frac{200}{1000}}{\frac{100 + 200}{1000}}$$

$$= \frac{0.6 \times 0.1 + 0.3 \times 0.2}{0.3} = \frac{0.06 + 0.06}{0.3}$$

$$= \frac{0.12}{0.3} = 0.4N$$

28. (3)

$$\Delta S_{\text{mix}} > 0$$

29. (2)

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{mass of solvent (in kg)}}$$

30. (2)

$$0.75 = \frac{w}{106} \times \frac{1000}{500}$$

$$w = 39.75\text{g}$$

31. (1)

Total boiling occurs at lower temperature if atmospheric pressure is lower than 76 cm Hg

32. (1)

$$P_M = 90 \times \frac{2}{5} + 70 \times \frac{3}{5}$$

$$= 78 \text{ torr}$$

33. (1)

Elevated boiling point means vapour pressure of the solution is lower than expected. Hence, negative deviation

34. (3)

$$P'_A = P_A^0 \cdot X_A = P_M \cdot Y_A$$

$$P'_A = P_B^0 \cdot X_B = P_M \cdot Y_B$$

$$\therefore \frac{P_A^0}{P_B^0} \cdot \frac{X_A}{X_B} = \frac{Y_A}{Y_B}$$

$$\therefore \frac{P_A^0}{P_B^0} > 1 \quad \therefore \frac{X_A}{X_B} < \frac{Y_A}{Y_B}$$

35. (1)

The interparticle forces in between CHCl_3 and acetone increase due to H-bonding and thus $\Delta_{\text{mixing}} V$ becomes negative

36. (2)

$$P_{\text{Total}} = P_A^0 X_A + P_B^0 X_B$$

$$= P_A^0 X_A + P_B^0 (1 - X_A)$$

$$= P_B^0 - (P_B^0 - P_A^0) X_A$$

$$\text{Given eq.} = 235 - 124 X_A$$

Compare with the given equation,

$$\therefore P_B^0 = 235, P_B^0 - P_A^0 = 124$$

$$\therefore P_A^0 = 235 - 124 = 111$$

37. (1)

The definition of azeotropic mixture

38. (3)

In cooker, due to increase in surrounding pressure, BP temperature also increases.

39. (3)

$$M = \frac{1000 \times k_f \times w}{\Delta T \times W}$$

$$62 = \frac{1000 \times 1.86 \times 50}{18.6 \times W}$$

$$W = 80.65 \text{ g}$$

$$\text{Total water} = 200 \text{ g}$$

$$\text{Hence, ice separated} = (200 - 80.65) \text{ g} = 119.35$$

40. (4)

Elevation in boiling point is a colligative property as it depends upon the number of particles

$$\Delta T_b \propto n$$

$$\text{For KCl } n = 2 \quad \Delta T_b = 0.2^\circ\text{C}$$

$$\text{For } \text{K}_4[\text{Fe}(\text{CN})_6], n = 5, \Delta T_b = 0.5^\circ\text{C}$$

41. (1)

When raisins are placed in salt water, they slowly get shrunk due to osmosis

42. (3)

$$\frac{p^0 - p_s}{p^0} = \frac{w \times M}{m \times W}$$

$$\frac{142 - p_s}{142} = \frac{0.4}{65} \times \frac{154}{1.6 \times 100}$$

[$\therefore$ molecular weight of $\text{CCl}_4 = 154$ and weight = density $\times$ volume]

$$142 - P_s = 0.841$$

$$P_s = 141.16 \text{ mm}$$

43. (2)

$$P_A = 0.1 \times 2 \times ST = 2P$$

$$P_B = 0.05 \times 3 \times ST = 1.5 P$$

44. (4)

According to Henry's law, the gas in contact with the liquid should not cause any chemical change.

45. (1)

(A $\rightarrow$ Q)

Hypertonic solutions have higher osmotic pressure than some other solutions

(B $\rightarrow$ P)

Isotonic solutions have same osmotic pressure than some other solutions

(C $\rightarrow$ P)

(D $\rightarrow$ S)

46. (2)

$$\text{Molality} = \frac{\text{Mole of solute}}{\text{wt. of water in kg}} = \frac{18 \times 1000}{180 \times 500} = 0.2 \text{ m}$$

47. (4)

Osmotic pressure (π) = CRT

$$\pi_1 = \pi_2 \text{ or } C_1 = C_2$$

$$\frac{q}{180 \times 0.5} = \frac{8}{180 \times 1}$$

$$w = 4 \text{ g}$$

48. (200)

$$w = 0.3 \text{ g}, W = 30 \text{ g},$$

$$\Delta T_b = 0.108^\circ\text{C}$$

$$k_b = 2.16^\circ\text{C}$$

$$\therefore M = \frac{k_b \times w \times 100}{\Delta T_b \times W}$$

$$= \frac{2.16 \times 0.3 \times 1000}{0.108 \times 30} = 200$$

49. (75)

For two solutions to be isotonic

$$\pi_{\text{K}_2\text{SO}_4} = \pi_{\text{urea}}$$

$$C_1 RT \times (1 + 2\alpha) = C_2 RT$$

$$0.004 \times (1 + 2\alpha) = 0.01$$

$$\text{Hence, } \alpha = 0.75 \text{ or } 75\%$$

50. (1)

Given, mass of solvent (w) = 100 g

Depression in freezing point (ΔT_f) = $T^\circ\text{C}$

$$k_f = 8.0$$

$$\frac{\text{Mass of solute (w)}}{\text{Molecular mass of solute (M)}} = \frac{2.4}{32}$$

$$\Delta T_f = \frac{1000 \times k_f}{n \times w} \left(\frac{w}{M} \right)$$

$$\Delta T_f = \frac{1000 \times 8.0 \times 0.075}{6 \times 100}$$

SECTION-III (MATHEMATICS)**51. (1)**

$$\text{Given } \begin{vmatrix} x & 2 & -1 \\ 2 & 5 & x \\ -1 & 2 & x \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x & 2 & -1 \\ 2 & 5 & x \\ -3 & -3 & 0 \end{vmatrix} = 0 \quad [R_3 \rightarrow R_3 - R_2]$$

$$\Rightarrow -1(-6+15) - x[-3x+6] = 0$$

$$\Rightarrow x^2 - 2x - 3 = 0$$

$$\Rightarrow x = 3, -1$$

52. (2)

$$\text{Given, } f(\alpha) = \begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix}$$

$$= 1(\alpha^3 - 1) - \alpha(\alpha^2 - \alpha^2) + \alpha^2(\alpha - \alpha^4)$$

$$= \alpha^3 - 1 - 0 + \alpha^3 - \alpha^6$$

$$\Rightarrow f(\sqrt[3]{2})$$

$$= 2 - 1 + 2 - 4$$

$$= -1$$

53. (1)

$$\text{Given, } \Delta = \begin{vmatrix} (e^\alpha + e^{-\alpha})^2 & (e^\alpha - e^{-\alpha})^2 & 4^2 \alpha \beta \gamma \\ (e^\beta + e^{-\beta})^2 & (e^\beta - e^{-\beta})^2 & 4^2 \alpha \beta \gamma \\ (e^\gamma + e^{-\gamma})^2 & (e^\gamma - e^{-\gamma})^2 & 4^2 \alpha \beta \gamma \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_2$

$$= \begin{vmatrix} 4 & (e^\alpha - e^{-\alpha})^2 & 4^2 \alpha \beta \gamma \\ 4 & (e^\beta - e^{-\beta})^2 & 4^2 \alpha \beta \gamma \\ 4 & (e^\gamma - e^{-\gamma})^2 & 4^2 \alpha \beta \gamma \end{vmatrix}$$

 $= 0$ ($\because$ two columns are proportional)Hence, it is independent of α, β and γ .**54. (1)**

$$\Delta_1 = -\Delta_2$$

55. (3)

$$\text{Let } \Delta = \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & b+c & -2c \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_3$ and $C_2 \rightarrow C_2 + C_3$

$$\Delta = \begin{vmatrix} -a+c & 2a+b+c & a+c \\ 2b+a+c & -b+c & b+c \\ a-c & b-c & -2c \end{vmatrix}$$

Now, applying $R_1 \rightarrow R_1 + R_3$ and $R_2 \rightarrow R_2 + R_3$

$$\Delta = \begin{vmatrix} 0 & 2(a+b) & a-c \\ 2(a+b) & 0 & b-c \\ a-c & b-c & -2c \end{vmatrix}$$

On expanding, we get

$$\Delta = -2(a+b) \{-2c[2(a+b)] - (a-c)(b-c)\} + (a-c)[2(a+b)(b-c)]$$

$$\begin{aligned} \Delta &= 8c(a+b)(a+b) + 4(a+b)(a-c)(b-c) \\ &= 4(a+b)[2ac + 2bc + ab - bc - ac + c^2] \\ &= 4(a+b)[ac + bc + ab + c^2] \\ &= 4(a+b)[c(a+c) + b(a+c)] \\ &= 4(a+b)(b+c)(c+a) \\ &= 2\alpha(a+b)(b+c)(c+a) \end{aligned}$$

Hence, $\alpha = 2$ **56. (2)**

$$\frac{1}{2} x^2 y^2 z^2 \begin{vmatrix} 0 & x & x \\ y & 0 & y \\ z & z & 0 \end{vmatrix}$$

$$= \frac{1}{2} x^2 y^2 z^2 (xyz + xyz)$$

$$= x^3 y^3 z^3$$

57. (3)

$$\therefore \begin{vmatrix} 3 & 1 & 4 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix} = 3(-2) + 1(4-0) + 4(1-0) = 2$$

58. (3)

$$\begin{aligned} \Delta &= 1(\sin\theta) - 1(-\cos\theta) + \\ &1(1 - (1+\sin\theta)(1+\cos\theta)) \\ &= \sin\theta + \cos\theta + (1 - \sin\theta - \cos\theta - 1 - \sin\theta \cos\theta) \\ &= -\sin\theta \cos\theta \end{aligned}$$

59. (2)

$$\begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix} = \begin{vmatrix} -2y & y & y \\ y & -2y & y \\ x+y & x+2y & x \end{vmatrix} \begin{vmatrix} R_1 - R_2 \\ R_2 - R_3 \end{vmatrix}$$

$$= y^2 \begin{vmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ x+y & x+2y & x \end{vmatrix} = y^2 \{(-2)[-2x - x - 2y]$$

$$- (x - x - y) + (x + 2y + 2x + 2y)\} \\ = y^2 \{6x + 4y + y + 3x + 4y\} = 9y^2(x + y)$$

60. (2)

We have,

$$\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = k(x+y+z)(x-z)^2$$

$$\text{LHS} = (x+y+z) \begin{vmatrix} 2 & 1 & 1 \\ z+x & z & x \\ x+y & y & z \end{vmatrix} \quad (R_1 \rightarrow R_1 + R_2 + R_3)$$

$$= (x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ x & z & x \\ x & y & z \end{vmatrix}$$

$$= (x+y+z) \{ 1(z^2 - xy) - 1(xz - x^2) + 1(xy - xz) \}$$

$$= (x+y+z)(x^2 + z^2 - 2xz)$$

$$\Rightarrow (x+y+z)(x-z)^2 = k(x+y+z)(x-z)^2 \quad (\text{given})$$

$$\Rightarrow k = 1$$

61. (1)

$$\text{Given, } \begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$$

$$\Rightarrow x(x^2 - 12) - 3(2x - 14) + 7(12 - 7x) = 0$$

$$\Rightarrow x^3 - 67x + 126 = 0$$

$$\Rightarrow (x+9)(x^2 - 9x + 14) = 0$$

$$\Rightarrow (x+9)(x-2)(x-7) = 0$$

$$\Rightarrow x = -9, 2, 7$$

Hence, the other two roots are 2, 7

62. (1)

We have,

$$\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} p & b & c \\ a-p & q-b & 0 \\ 0 & b-q & r-c \end{vmatrix} = 0$$

$$\left[\begin{array}{l} \text{Applying } R_3 \rightarrow R_3 - R_2 \\ \text{and } R_2 \rightarrow R_2 - R_1 \end{array} \right]$$

$$\Rightarrow \begin{vmatrix} p & b & c \\ p-a & q-b & r-c \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{p}{p-a} + \frac{b}{q-b} + \frac{c}{r-c} = 0$$

$$\Rightarrow \frac{p}{p-a} + \left(\frac{q}{q-b} - 1 \right) + \left(\frac{r}{r-c} - 1 \right) = 0$$

$$\Rightarrow \frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = 2$$

63. (2)

$$\begin{vmatrix} a & b-y & c-z \\ a-x & b & c-z \\ a-x & b-y & c \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a & b-y & c-z \\ -x & y & 0 \\ 0 & -y & z \end{vmatrix} = 0$$

(Using $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_2$)

$$\Rightarrow a(yz) + x(bz - yz + cy - yz) = 0$$

$$\Rightarrow ayz + bzx + cyx = 2xyz$$

$$\Rightarrow \frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$$

64. (4)

$$\begin{vmatrix} \log_5 729 & \log_3 5 \\ \log_5 27 & \log_9 25 \end{vmatrix} = \begin{vmatrix} \log_5 3^6 & \log_3 5 \\ \log_5 3^3 & \log_{3^2} 5^2 \end{vmatrix}$$

$$= \begin{vmatrix} 6\log_5 3 & \log_3 5 \\ 3\log_5 3 & \frac{2}{2}\log_3 5 \end{vmatrix}$$

$$= 6\log_5 3 \log_3 5 - 3\log_5 3 \log_3 5$$

$$= 6 - 3 = 3$$

$$\text{And } \begin{vmatrix} \log_3 5 & \log_{27} 5 \\ \log_5 9 & \log_5 9 \end{vmatrix} = \begin{vmatrix} \log_3 5 & \log_{3^3} 5 \\ \log_5 3^2 & \log_5 3^2 \end{vmatrix}$$

$$= \begin{vmatrix} \log_3 5 & \frac{1}{3}\log_3 5 \\ 2\log_5 3 & 2\log_5 3 \end{vmatrix}$$

$$= 2\log_5 3 \log_3 5 - \frac{2}{3}\log_5 3 \log_3 5$$

$$= 2 - \frac{2}{3} = \frac{4}{3}$$

Now,

$$\begin{vmatrix} \log_5 729 & \log_3 5 \\ \log_5 27 & \log_9 25 \end{vmatrix} \begin{vmatrix} \log_3 5 & \log_{27} 5 \\ \log_5 9 & \log_5 9 \end{vmatrix} = 3 \cdot \frac{4}{3} = 4$$

Take option (4),

$$\log_3 5 \cdot \log_5 81 = \log_3 81 = \log_3 3^4 = 4$$

65. (2)

$$\sum \Delta_k = \begin{vmatrix} \frac{n(n+1)}{2} & 1 & 5 \\ \frac{n(n+1)(2n+1)}{6} & 2n+1 & 2n+1 \\ \frac{n^2(n+1)^2}{4} & 3n^2 & 3n+1 \end{vmatrix}$$

$$\begin{aligned}
&= \frac{n(n+1)(2n+1)}{2} \begin{vmatrix} 1 & 1 & 5 \\ \frac{1}{3} & 1 & 1 \\ \frac{n(n+1)}{2} & 3n^2 & 3n+1 \end{vmatrix} \\
&= \frac{n(n+1)(2n+1)}{2} \left[3n+1-3n^2 - \left(n + \frac{1}{3} - \frac{n^2}{2} - \frac{n}{2} \right) + 5 \left(n^2 - \frac{n^2}{2} - \frac{n}{2} \right) \right] \\
&= \frac{n(n+1)(2n+1)}{2} \left[3n+1-3n^2 - \frac{n}{2} - \frac{1}{3} + \frac{n^2}{2} + \frac{5n^2}{2} - \frac{5n}{2} \right] \\
&= \frac{n(n+1)(2n+1)}{2} \left[\frac{2}{3} \right] = 2 \sum k^2
\end{aligned}$$

66. (2)

$$\text{Given } \begin{vmatrix} x+1 & 2x+1 & 3x+1 \\ 2x & 4x+3 & 6x+3 \\ 4x+4 & 6x+4 & 8x+4 \end{vmatrix} = 0$$

$$\Rightarrow 2 \begin{vmatrix} 0 & x & 2x \\ 2x & 4x+3 & 6x+3 \\ 2x+2 & 3x+2 & 4x+2 \end{vmatrix} = 0$$

[Using $(R_1 \rightarrow 2R_1 - R_3)$]

$$\Rightarrow 2 \begin{vmatrix} 0 & x & 0 \\ 2x & 4x+3 & -2x-3 \\ 2x+2 & 3x+2 & -2x-2 \end{vmatrix} = 0$$

[Using $(C_3 \rightarrow C_3 - 2C_2)$]

$$\Rightarrow -4x[2x^2 + 2x - (2x+3)(x+1)] = 0$$

$$\Rightarrow -4x[2x^2 + 2x - (2x^2 + 5x + 3)] = 0$$

$$\Rightarrow 4x(3x+3) = 0$$

$$\Rightarrow x+1=0 \quad [\because x \neq 0 \text{ given}]$$

67. (4)

We have,

$$\begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$$

$$= \begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}^2$$

$$\begin{aligned}
&\text{Now, } \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \\
&= \begin{vmatrix} 1 & 1 & 1 \\ 0 & \alpha-1 & \beta-1 \\ 0 & \alpha^2-1 & \beta^2-1 \end{vmatrix} \quad \left[\begin{array}{l} \text{Applying } R_2 \rightarrow R_2 - R_1, \\ R_3 \rightarrow R_3 - R_1 \end{array} \right] \\
&= (\alpha-1)(\beta^2-1) - (\beta-1)(\alpha^2-1) \\
&= \alpha\beta^2 - \alpha - \beta^2 - \alpha^2\beta + \beta + \alpha^2 \\
&= (\alpha^2 - \beta^2) - (\alpha - \beta) - \alpha\beta(\alpha - \beta) \\
&= (\alpha - \beta)[\alpha + \beta - 1 - \alpha\beta] \\
&= \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} \{ \alpha + \beta - 1 - \alpha\beta \} \\
&= \sqrt{\frac{b^2 - 4ac}{a^2}} \left\{ -\frac{b}{a} - 1 - \frac{c}{a} \right\} \\
&= -\sqrt{\frac{b^2 - 4ac}{a^2}} \left(\frac{a+b+c}{a} \right)
\end{aligned}$$

$$\text{Hence, } \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$$

$$\begin{aligned}
&= \left\{ -\sqrt{\frac{b^2 - 4ac}{a^2}} \left(\frac{a+b+c}{a} \right) \right\}^2 \\
&= \frac{(b^2 - 4ac)(a+b+c)^2}{a^4}
\end{aligned}$$

68. (3)

Given, $a_1, a_2, a_3, \dots \in \text{GP}$

$\Rightarrow \log a_1, \log a_2, \dots \in \text{AP}$

$\Rightarrow \log a_n, \log a_{n+1}, \log a_{n+2}, \dots \in \text{AP}$

$$\Rightarrow \log a_{n+1} = \frac{\log a_n + \log a_{n+2}}{2} \quad \dots(i)$$

$$\text{Similarly, } \log a_{n+4} = \frac{\log a_{n+3} + \log a_{n+5}}{2} \quad \dots(ii)$$

$$\text{and } \log a_{n+7} = \frac{\log a_{n+6} + \log a_{n+8}}{2} \quad \dots(ii)$$

$$\text{Given, } \Delta = \begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$$

$$\text{Applying } C_2 \rightarrow C_2 - \frac{C_1 + C_3}{2}$$

$$\Delta = \begin{vmatrix} \log a_n & 0 & \log a_{n+2} \\ \log a_{n+3} & 0 & \log a_{n+5} \\ \log a_{n+6} & 0 & \log a_{n+8} \end{vmatrix} = 0$$

69. (4)

Use expansion $\lambda = 4$

$$(\sqrt{2})^{\log_2(4)^2} = 4$$

70. (1)

$$\text{Let } A \equiv \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 5 \quad \dots(i)$$

$$\therefore \begin{vmatrix} b_2c_3 - b_3c_2 & c_2a_3 - c_3a_2 & c_2b_3 - c_3b_2 \\ b_3c_1 - b_1c_3 & c_3a_1 - c_1a_3 & a_3b_1 - a_1b_3 \\ b_1c_2 - b_2c_1 & c_1a_2 - c_2a_1 & a_1b_2 - a_2b_1 \end{vmatrix}$$

$$|\text{adj } A| = (5)^{3-1} \quad [\text{from Eq. (i)}]$$

$$= 5^2 = 25 \quad \left(\because |\text{adj } A| = |A|^{n-1} \right)$$

71. (1)

$$1(-10x^2 - 10x) - 4(5x^2 - 5) + 20(2x + 2)$$

$$= -10x^2 - 10x - 20x^2 + 20 + 40x + 40$$

$$= -30x^2 + 30x + 60$$

sum of real root is 1.

72. (0)

$$\begin{vmatrix} 5^2 & 5^3 & 5^4 \\ 5^3 & 5^4 & 5^5 \\ 5^4 & 5^5 & 5^6 \end{vmatrix}$$

$$= 5^2 \times 5^3 \times 5^4 \begin{vmatrix} 1 & 5 & 5^2 \\ 1 & 5 & 5^2 \\ 1 & 5 & 5^2 \end{vmatrix}$$

$$= 5^2 \times 5^3 \times 5^4 \times 5 \begin{vmatrix} 1 & 1 & 5^2 \\ 1 & 1 & 5^2 \\ 1 & 1 & 5^2 \end{vmatrix}$$

$$= 5^2 \times 5^3 \times 5^4 \times 0$$

$$= 0$$

73. (0)

$$\begin{vmatrix} \log e & \log e^2 & \log e^3 \\ \log e^2 & \log e^3 & \log e^4 \\ \log e^3 & \log e^4 & \log e^5 \end{vmatrix}$$

$$= \begin{vmatrix} \log e & 2\log e & 3\log e \\ 2\log e & 3\log e & 4\log e \\ 3\log e & 4\log e & 5\log e \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

74. (1)

Given,

$$\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = (y-z)(z-x)(x-y) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$$

The degree of determinant

$$= n + (n+2) + (n+3) = 3n+5$$

And the degree of RHS = 2

$$\therefore 3n+5 = 2 \Rightarrow n = -1$$

75. (1)

$$\text{LHS} = \begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$$

The determinant can be written sum of $2 \times 2 \times 2 = 8$ determinants of which 6 reduces to zero because their two rows are identical.

$$\therefore \text{RHS} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

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